

Nachiket Kerai

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Looking to expand my industry experience through internship opportunities in Embedded Systems, Firmware and Systems Engineering.

Education

University of Wisconsin-Madison

Madison, WI

B.S. Computer Engineering and Computer Science

December 2025

- **Courses:** Programming (I, II, and III), Signals Information and Computing, Digital System Design and Synthesis, Intro to Algorithms, Machine Organization and Programming, Electrodynamics, Database Management Systems, Operating Systems, Microprocessor Systems, Circuit Analysis, Intro to Artificial Intelligence
- **Awards:** Dean's Honor List Fall 2022, Spring 2023, Fall 2023, Spring 2024, Fall 2024
- GPA: (3.72/4)

Experience

Visual Computing Lab @ UW-Madison

Madison, WI

Undergraduate Researcher

January, 2025 – Present

- ◇ Developed Python programs to process time-of-flight sensor data into computer vision datasets.
- ◇ Working on robot arm movement through enhancing robot kinematics to capture sensor data.

Bick Digital Entomology Lab @ UW-Madison

Madison, WI

Embedded Systems Research Intern

May, 2024 – Present

- ◇ Designed and created a 2D depth imaging device with an infrared time-of-flight sensor, ESP-32 Microcontroller and Raspberry Pi for insect biomarker classification using Arduino code (C++) and Python.
- ◇ Tested device to find sensor limitations and optimal environments for depth measurements.

Computational Optics Group @ UW-Madison

Madison, WI

Computational Optics Undergraduate Research Assistant

February, 2023 – December, 2023

- ◇ Developed and optimized Python image reconstruction algorithms, reducing reconstruction time by 20%.
- ◇ Worked with C++ and MATLAB non line of sight image reconstruction algorithms.
- ◇ Performed experiments using LIDAR data to determine factors that affect image signal to noise ratio.

Projects

Press-your-Luck Embedded Game

- ◇ Programmed all device drivers and firmware for LCD, buttons, and joystick on an ARM Cortex-M microcontroller.
- ◇ Designed and programmed game mechanics such as score updating, item selection, and spin management.
- ◇ (C, freeRTOS, Arm Cortex-M, I2C, UART)

Unix Shell

- ◇ Developed a Unix shell with interactive and batch modes, built-in commands, process management, redirection, variable substitution, and command history.
- ◇ Implemented custom parsing, error handling, and memory management to ensure robust user functionality.
- ◇ (C, UNIX Programming)

Verilog Combination Alarm Clock

- ◇ Created an alarm clock on an FPGA with basic functionalities such as a timer, alarm, and stopwatch.
- ◇ (Verilog, Quartus Prime, FPGAs)

Technical Skills

Skills: C, Python, MATLAB, Verilog, Object Oriented Programming, Data Structures, Algorithms, freeRTOS, UNIX

Tools: Git, Bash, Quartus Prime, ModelSim, LTSpice